Chapter 6. Permutation Tests for Analysis of Variance

Kruskal–Wallis, ordered alternatives, Friedman test, general two-way layouts

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6.1 One-way layout

Motivation. The Kruskal–Wallis test compares the average pooled ranks of k groups, a rank version of one-way ANOVA.

Observations $X_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$, with i.i.d. errors. We test $H_0 : \mu_1 = \cdots = \mu_k$.

The F-test. With group means $\bar{X}_i$ and grand mean $\bar{X}$,

$$SS_{\text{Tr}} = \sum_i n_i(\bar{X}_i - \bar{X})^2, \qquad SS_E = \sum_{i,j}(X_{ij} - \bar{X}_i)^2, \qquad F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/(n-k)}.$$

Proposition 6.1

$SS_{\text{Tot}} := \sum_{i,j}(X_{ij} - \bar{X})^2 = SS_{\text{Tr}} + SS_E$. Under H_0 with normal errors, $F \sim F_{k-1,n-k}$.

Proof [Supplement]. Expand $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$; the cross term $2\sum_i(\bar{X}_i - \bar{X})\sum_j(X_{ij} - \bar{X}_i) = 0$. Under normality $SS_E/\sigma^2 \sim \chi^2_{n-k}$ independently of the group means, and SS_{Tr}/σ^2 is the squared projection of the standard normal $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ onto the complement of $(\sqrt{n_i})_i$, so χ^2_{k-1}. ■

Permutation F-test. Under H_0 (equal distributions) the n observations are exchangeable: relabel among the $n!/\prod n_i!$ equally likely splits. With SS_{Tot} fixed, $F = \frac{n-k}{k-1} \cdot \frac{SS_{\text{Tr}}}{SS_{\text{Tot}} - SS_{\text{Tr}}}$ is increasing in SS_{Tr}. For moderate samples the permutation law of F is close to $F_{k-1,n-k}$ without normality.

6.1.1 The Kruskal–Wallis test

Rank jointly; $\bar{R}_i$ is the mean rank of group i, S_R^2 the sample variance of all ranks ($n(n+1)/12$ without ties).

Definition (Kruskal–Wallis statistic)

$KW = \frac{1}{S_R^2}\sum_{i=1}^k n_i\Big(\bar{R}_i - \frac{n+1}{2}\Big)^2$, which without ties equals $\frac{12}{n(n+1)}\sum_i n_i(\bar{R}_i - \frac{n+1}{2})^2$.

Proposition 6.2

(i) The permutation test based on KW coincides with the permutation F-test applied to the ranks.
(ii) Under H_0, $KW \xrightarrow{d} \chi^2_{k-1}$ as all $n_i \to \infty$.

Proof. (i) For ranks, $SS^R_{\text{Tot}} = (n-1)S_R^2$ is fixed and $SS^R_{\text{Tr}} = S_R^2\, KW$, so $F_R = \frac{n-k}{k-1}\frac{KW}{n-1-KW}$ is increasing in KW.

(ii) With $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$, Theorem 4.2 gives $\text{Cov}(Z) \to I - \mathbf{u}\mathbf{u}'$, $\mathbf{u} = (\sqrt{n_i/n})_i$, a rank-$(k-1)$ projection; a multivariate finite-population CLT gives $Z \xrightarrow{d} N(0, I - \mathbf{u}\mathbf{u}')$, so $KW = \|Z\|^2 \xrightarrow{d} \chi^2_{k-1}$. ■

In small samples prefer the exact or Monte Carlo permutation law to χ^2.

6.1.2 Ordered alternatives

Against $H_1 : \mu_1 \le \mu_2 \le \cdots \le \mu_k$ (at least one strict), the **Jonckheere–Terpstra** statistic sums the pairwise Mann–Whitney counts:

$$JT = \sum_{i<l} U_{il}, \qquad U_{il} = \#\{(a,b) : X_{ia} < X_{lb}\}.$$

Large JT supports the trend. Up to an affine map it is Kendall's τ between observations and group index; its permutation law gives an exact test, more powerful than KW against monotone trends.

6.2 Randomized complete block design

k treatments in each of b blocks, one observation per cell: $X_{ij} = \mu + t_i + \beta_j + \epsilon_{ij}$. We test $H_0 : t_1 = \cdots = t_k$.

Classical F. $SS_{\text{Tr}} = b\sum_i(\bar{X}_{i\cdot} - \bar{X})^2$ and $SS_E = \sum_{i,j}(X_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X})^2$. Under normality, $F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/((k-1)(b-1))} \sim F_{k-1,(k-1)(b-1)}$.

Why shuffle within blocks. Under H_0, $X_{ij} = \mu+\beta_j+\epsilon_{ij}$: observations are exchangeable within, not across, blocks. So permute labels independently within blocks, $(k!)^b$ arrangements. These leave block means and SS_{Tot} unchanged, so $SS_E = SS_{\text{Tot}} - SS_{\text{Blocks}} - SS_{\text{Tr}}$ and the permutation F-test is the permutation test of SS_{Tr}.

6.2.1 The Friedman test

Rank within blocks; $R_i = \sum_j R_{ij}$, $\bar{R}_{i\cdot} = R_i/b$.

Definition (Friedman statistic)

$$FM = \frac{12b}{k(k+1)} \sum_{i=1}^{k} \left(\bar{R}_{i\cdot} - \frac{k+1}{2}\right)^2 = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) \text{ (no ties)}.$$

Under H_0 the within-block rank vectors are independent uniform permutations, so the null law is distribution-free and $FM \xrightarrow{d} \chi^2_{k-1}$ as $b \to \infty$. With ties divide by the average within-block rank variance instead of $k(k+1)/12$.

Example 6.1

$k = 3$, $b = 4$, ranks of (A, B, C): $(1,2,3)$, $(1,3,2)$, $(1,2,3)$, $(2,1,3)$. Rank sums $5, 8, 11$ give $FM = \frac{12}{48}(25 + 64 + 121) - 48 = 4.5$: χ^2_2 gives $p = e^{-2.25} = 0.105$, exact enumeration of $(3!)^4 = 1296$ arrangements $p = 162/1296 = 0.125$.

6.2.2 Page's trend test

For $H_1 : t_1 \le \cdots \le t_k$ use $L = \sum_{i=1}^{k} iR_i$, a block sum of Spearman-type statistics.

Proposition 6.3

Under H_0, $E(L) = \dfrac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \dfrac{bk^2(k+1)(k^2-1)}{144}$.

Proof. In block j, $L_j = \sum_i iR_{ij}$ with a uniform rank permutation, $ER_{ij} = \frac{k+1}{2}$, so $EL_j = \frac{k+1}{2} \cdot \frac{k(k+1)}{2}$. Proposition 5.2 with $a_i = i$ gives $\mathrm{Var}\, L_j = \frac{1}{k-1}\Big(\sum_i (i - \frac{k+1}{2})^2\Big)^2 = \frac{1}{k-1}\Big(\frac{k(k^2-1)}{12}\Big)^2 = \frac{k^2(k+1)(k^2-1)}{144}$. Blocks are independent; multiply by b. ■

6.3 General two-way layouts

With c replicates per cell and no interaction, $X_{ijl} = \mu + t_i + \beta_j + \epsilon_{ijl}$; permute all kc observations within blocks. The rank version (**Skillings–Mack**) uses

$$SM = \frac{b\sum_i (\bar{R}_{i\cdot} - \bar{R})^2}{\text{average within-block rank variance}} \dot{\sim} \chi^2_{k-1}.$$

Prefer the permutation law in small designs. With interaction, ranking can destroy or create it: use aligned-rank procedures or the regression resampling of Chapter 10.

6.4 Selected exercises

Motivation. Friedman as the sign test and as Cochran's Q, and Page's moments with ties.

Exercise 6.5 (Friedman with two treatments is the sign test)

In a complete block design with $k = 2$ treatments, b blocks and no ties within blocks, show that the permutation test based on the Friedman statistic FM is the two-sided sign test applied to the differences $D_j = X_{1j} - X_{2j}$.

Solution.

Step 1. Let $S = \#\{j : D_j > 0\}$. Within block j treatment 1 has rank $R_{1j} = 1 + I(D_j > 0)$, so $R_1 = b + S$ and $R_2 = 3b - R_1 = 2b - S$.

Step 2. By the second form of the definition with $k = 2$,

$$FM = \frac{2}{b}\{(b+S)^2 + (2b-S)^2\} - 9b = \frac{2}{b}(5b^2 - 2bS + 2S^2) - 9b$$
$$= \frac{(2S-b)^2}{b} = \Big(\frac{S - b/2}{\sqrt{b/4}}\Big)^2.$$

So FM is the squared standardized sign statistic, increasing in $|S - b/2|$.

Step 3. Within-block permutations swap the labels independently with probability $\frac{1}{2}$, flipping $\operatorname{sgn} D_j$; so $S \sim B(b, \frac{1}{2})$ (Proposition 2.1), and large FM means large $|S - b/2|$: the two-sided sign test, with χ^2_1 as its normal approximation.

So for $k = 2$ and normal errors its ARE relative to the paired t-test is $2/\pi$ (Section 2.3); the general value $3k/(\pi(k+1))$ (not proved here) increases toward $3/\pi$. ■

Exercise 6.6 (Page's L with ties)

In a complete block design rank within blocks using mid-ranks, and let S_j^2 be the sample variance (divisor $k-1$) of the ranks in block j. Show that under H_0

$$E(L) = \frac{bk(k+1)^2}{4}, \qquad \mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12}\sum_{j=1}^{b} S_j^2,$$

and that this reduces to Proposition 6.3 without ties.

Solution.

Step 1. Under H_0, given the multiset of mid-ranks in block j, $(R_{1j}, \ldots, R_{kj})$ is a uniform permutation of it; blocks are independent.

Step 2. Mid-ranks keep the total $k(k+1)/2$, so $ER_{ij} = \frac{k+1}{2}$ and $E(L_j) = \frac{k+1}{2}\sum_i i = \frac{k(k+1)^2}{4}$, $L_j = \sum_i i R_{ij}$.

Step 3. The proof of Proposition 5.2 needs only fixed $b_1, \ldots, b_k$ permuted against fixed $a_1, \ldots, a_k$: $\mathrm{Var}\left(\sum_i a_i b_{\pi(i)}\right) = S_{aa}S_{bb}/(k-1)$, $S_{aa} = \sum_i (a_i - \bar{a})^2$. With $a_i = i$ ($S_{aa} = k(k^2-1)/12$) and the mid-ranks of block j ($S_{bb} = (k-1)S_j^2$),

$$\mathrm{Var}(L_j) = \frac{k(k^2-1)}{12} S_j^2.$$

Step 4. Sum over independent blocks. Without ties $S_j^2 = k(k+1)/12$ and $\mathrm{Var}(L) = \frac{k(k-1)(k+1)}{12} \cdot \frac{bk(k+1)}{12} = \frac{bk^2(k+1)(k^2-1)}{144}$. ■

Exercise 6.7 (Binary responses: Cochran's Q)

Let $x_{ij} \in \{0, 1\}$ (success or failure of treatment i in block j), with treatment totals T_i, block totals B_j and grand total T. The tie-corrected Friedman statistic is $FM = b\sum_i (\bar{R}_{i\cdot} - \frac{k+1}{2})^2 / (\frac{1}{b}\sum_j S_j^2)$, with mid-ranks and S_j^2 as in Exercise 6.6.
(a) Show that FM equals Cochran's statistic

$$Q = \frac{(k-1)\left(k\sum_i T_i^2 - T^2\right)}{kT - \sum_j B_j^2}.$$

(b) Four treatments of plantar warts were applied to each of 7 patients (blocks), with outcomes (1 = success):

	P1	P2	P3	P4	P5	P6	P7
Freezing	0	1	0	0	0	1	0
Surgery	1	1	0	1	0	1	1
Topical medication	0	0	0	1	0	1	0
Duct tape	1	1	0	0	1	0	1

Test whether the treatments differ.

Solution.

(a)

Step 1. (Mid-ranks.) In block j the $k - B_j$ failures get $c_j = \frac{k - B_j + 1}{2}$ and the successes $\frac{2k - B_j + 1}{2} = c_j + \frac{k}{2}$, so $R_{ij} = c_j + \frac{k}{2}x_{ij}$ (also for $B_j \in \{0, k\}$).

Step 2. (Numerator.) $\bar R_{i\cdot} = \bar c + \frac{k}{2b}T_i$, whose average over i is $\frac{k+1}{2}$, so $\bar R_{i\cdot} - \frac{k+1}{2} = \frac{k}{2b}(T_i - T/k)$ and

$$b\sum_i \Big(\bar R_{i\cdot} - \frac{k+1}{2}\Big)^2 = \frac{k^2}{4b}\Big(\sum_i T_i^2 - \frac{T^2}{k}\Big).$$

Step 3. (Denominator.) $S_j^2 = \frac{k^2}{4}\cdot\frac{1}{k-1}\sum_i \big(x_{ij} - \frac{B_j}{k}\big)^2 = \frac{k^2}{4(k-1)}\Big(B_j - \frac{B_j^2}{k}\Big) = \frac{kB_j(k-B_j)}{4(k-1)}$.

Step 4. (Ratio.) Using $\sum_j B_j(k-B_j) = kT - \sum_j B_j^2$,

$$FM = \frac{\frac{k^2}{4b}\big(\sum_i T_i^2 - T^2/k\big)}{\frac{1}{b}\cdot\frac{k}{4(k-1)}\big(kT - \sum_j B_j^2\big)} = \frac{(k-1)\big(k\sum_i T_i^2 - T^2\big)}{kT - \sum_j B_j^2} = Q.$$

Blocks with $B_j \in \{0, k\}$ contribute to neither part: they carry no information.

(b) $T_i = (2,5,2,4)$, $T = 13$, $\sum T_i^2 = 49$; $B_j = (2,3,0,2,1,3,2)$, $\sum B_j^2 = 31$. So

$$Q = \frac{3(4\cdot 49 - 169)}{4\cdot 13 - 31} = \frac{81}{21} = 3.86.$$

The χ^2_3 approximation gives $p = 0.28$, while enumerating all 13,824 within-block arrangements gives the exact $p = 5148/13824 = 0.37$. No evidence of a difference; with six informative blocks χ^2 is noticeably off, as in Example 6.1. ■